

TEAMMATE GUIDE

CFM Coach

A learning loop around CFM Agent v5.0. It records every forecast the agent makes, waits for the world to resolve it, scores what happened, and proposes — never applies — calibration corrections.

Purpose

A WTI forecast that can be relied on for direction *and* risk, plus an explicit signal for when to distrust it.

Core principle

The coach measures and proposes. Every change waits for a human, and **cfm_agent_v_5_0 is never modified**.

How it learns

M1 calibration of the constants (built), M2 memory of the agent's own track record, M3a analogue lookup over 4,700 historical days.

State today

Recorder live and scheduled. M1 harness built and tested. One clean live record; first real evidence resolves 21 Aug 2026.

Prepared for internal team collaboration
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1 | EXECUTIVE OVERVIEW

What CFM Coach does

CFM Coach is a separate package that sits *around* CFM Agent v5.0. The agent produces a probabilistic WTI forecast; the coach records exactly what produced it, waits for the horizon to elapse, measures how good it was, and works out which constants should change. It is the measurement instrument the agent has never had.

The short version

v5.0 is a well-controlled pipeline with no feedback. It converts the LLM's categorical judgment into numbers using a set of fixed constants — action fractions, uncertainty multipliers, tier thresholds — and nothing has ever checked whether those constants are right. The coach closes that loop, slowly and conservatively, without touching the agent.

- **Record** every run with enough detail to reproduce it exactly, months later.
- **Resolve** each horizon against what WTI actually did.
- **Score** the agent against the raw ensemble and against candidate calibrations, on identical days.
- **Gate** any proposed change behind a locked statistical test.
- **Propose** the change to a human. Nothing is ever applied automatically.

The one-line boundary

The coach reads v5.0's outputs and injects settings at construction time. It never edits the agent package, so v5.0's manifest, its provenance claims, and its 43-file integrity check all stay true. Verified on every change.

What the coach produces

Output	What it contains
Run record	One JSON file per run (~90 KB): the LLM assessment, research packet, per-model and ensemble forecasts, policy decision, transformation audit, and the five identity fields needed to replay it — settings, calibration version, model, package fingerprint, prompt version.
Scoreboard	Pinball loss, CRPS, MAE, 80% interval coverage and mean width for three variants on identical days: the raw ensemble, the agent as published, and the calibration in force now.
Fidelity report	Proof that replaying each record under its own calibration reproduces its recorded quantiles exactly. If it ever fails, every fit built on that corpus is invalid.
Comparison verdict	Nine pass/fail conditions, a paired effect size, an origin-blocked bootstrap p-value, and the proposed value shrunk halfway toward the incumbent.
Trust report	A tier (high / medium / low / no_basis) plus one line naming the dominant driver. Built, and deliberately not wired in — see section 9.

2 | THE OBJECTIVE

Why the goal is "when to distrust it"

The instinct is to aim at accuracy. The arithmetic argues against making that the primary target. Improving accuracy means moving a *level* against a very noisy signal, and at a 21-day horizon there are only about four independent observations per year — even running daily, because overlapping windows are not independent. Detecting a real improvement that way takes years.

Knowing *when to distrust* the forecast is a different statistical object: a **ranking**. Were the days the system flagged as low-confidence actually worse than the days it did not? Rankings converge far faster than levels — months, not years. It is also the thing an LLM forecaster can do that ARIMA cannot: report what it looked for, what it found, what conflicted, and what it could not resolve.

The requirements that shape everything else

#	Requirement	Status
R1	Honest intervals: P10–P90 converging on 80% coverage.	0 of 12 — open
R2	A trust output, separate from the forecast.	Built, held back
R3	Cutoff-honest: no trust signal may depend on the outcome it assesses.	Enforced
R4	Trust validated by rank, or retired.	Blocked on corpus
R5	Out-of-distribution flagging. Annotates, never suppresses.	Built, weak
R6	Every run replayable.	Done and verified
R7	Only live forward-dated runs are fitting evidence.	Enforced by construction
R8	Nothing auto-applies to the agent.	Enforced
R9	Audit isolation preserved.	Enforced
R10	Improvement is a paired comparison on identical days.	Enforced by the gate
R11	Stage 1 is additive only — no existing file modified.	Holds

The architectural unlock

R2 — a trust output — turns out to be far cheaper than it sounds, because v5.0 already computes a great deal of trust-bearing signal and then discards it *by design*. Audit isolation correctly forbids these from influencing the forecast — but a trust score is not the forecast. A second output that consumes exactly what the forecast is forbidden to consume respects the architecture and recovers signal that is currently thrown away.

Signal v5.0 computes	Where it lives	Used for the forecast?
Evidence tier and tier level	policy_decision	Shapes caps, never reported
Material evidence conflict	llm_context_assessment	Disqualifies, then discarded
Source Validator findings	source_validation_audit	Audit-only
Claim-support entailment findings	claim_support_findings	Audit-only
Model disagreement std	numerical_suite_audit	Unused
Warnings, floor applied	forecast_transformation	Not surfaced
Market diagnostics (VIX, vol, contango)	numerical_suite_audit	Unused

This is the entire source of the stage-1 trust report

No case base is required for it, and nothing about it touches the forecast path. It is recovered signal, not new machinery — which is why R2 was scoped into stage 1 at all.

3 | ARCHITECTURE

Two loops, three learning mechanisms

The coach runs two workflows on very different clocks. One runs every business day and costs about four minutes. The other runs rarely, only when enough horizons have resolved, and involves no LLM at all.

Workflow A — the daily record (every business day)

1 Calibration Look up the version in force for today's cutoff.	2 Inject Build agent settings from that version and construct v5.0.	3 Run v5.0 executes unmodified: research, assessment, policy, engine.	4 Layer Apply the coach's post-engine centre/width layer.	5 Record Write one replayable RunRecord to the corpus.
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Every day produces a record whether or not anything is learned from it. **This loop is the corpus**, and it cannot be reconstructed later: re-running the agent at a past cutoff searches today's web, so a missed day is evidence permanently lost. The job is scheduled on weekdays at 08:00 local.

Workflow B — the reflection cycle (periodic, no LLM)

1 Resolve Join due horizons to realized WTI.	2 Verify Fidelity-check the corpus; stop if it drifted.	3 Replay Re-derive candidate and incumbent on the same days.	4 Score Pinball, CRPS, coverage.	5 Gate Nine locked conditions.	6 Propose Queue a shrunk change for human review.
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The three learning mechanisms

	Mechanism	Who learns	Changes the agent?	State
M1	Calibrate the constants that convert judgment into quantiles	The machinery	No	Built
M2	Memory of the agent's own track record	The LLM	Yes	Designed
M3a	Analogue lookup feeding a trust report	Nobody — it is a read	No	Built, held
M3b	Analogue injection into the prompt	The LLM	Yes	Deferred

Why M1 comes first

M1 is the measurement instrument for M2. M2 changes what the LLM says, and nothing else can tell you whether that helped. Without a paired harness you get what this repo's earlier adaptive agent got: CRPS 9.604 → 9.116 on n=22, with no standard error and no way to know whether anything actually happened.

4 | THE RECORDER

How replayability is achieved without touching v5.0

A calibration corpus needs five things per run that v5.0 does not persist: the settings it ran under, the calibration version they came from, the model string, a fingerprint of the agent package, and the prompt version. The insight that makes this free is that **the caller already knows all five before the run starts**.

So the coach constructs the settings itself, injects them through v5.0's own `build_cfm_agent_config(settings=...)` entry point, takes the predictions back, and pairs them with what it already held. The agent never learns the coach exists. The package fingerprint is a hash of v5.0's shipped manifest, so if the agent's constants ever change underneath the corpus, the affected records become visibly a different system.

Provenance: the field that protects every conclusion

Only runs issued on their own cutoff count as fitting evidence. A run re-executed at a past cutoff re-searches *today's* web, and even when every cited source predates the cutoff, the *selection* of what to look at is shaped by what turned out to matter. v5.0's leakage verifier inspects content, not retrieval, so it cannot catch this. The coach excludes such runs by construction rather than by discipline.

Provenance	Meaning	Fitting evidence?
live_forward	Issued on or after its own cutoff. Genuinely forward-looking.	Yes
replayed_live_search	Re-executed at a past cutoff against today's web.	No
backfill_cached_news	Reconstructed from cached research.	No
ensemble_only	Numerical suite with no LLM stage.	No

This is not a hypothetical distinction

Both audit files that existed before the coach was built turned out to have been issued **164 days after** their cutoff. They are recorded as replayed and excluded. Section 9 explains what that cost us.

5 | REPLAY AND CALIBRATION

M1: re-deriving a forecast without an LLM

The core of M1 is the ability to take a stored run and ask: *what would this same judgment have produced under different constants?* Answering that takes microseconds, offline, with no search and no tokens.

It is exact rather than approximate because of one property of v5.0: **the prompt contains no numeric constants**. The LLM emits categorical judgments — `large_up`, `substantially_wider` — and Python owns every number that turns those into quantiles. The LLM's output is therefore independent of the constants by construction, so re-deciding a stored assessment under new constants is a valid counterfactual, not a guess.

The arithmetic is not reimplemented. The coach *imports* v5.0's own `EvidencePolicy` and `PythonForecastEngine` and feeds them reconstructed inputs, so its model of the agent cannot drift from the agent.

The two levers

Lever	Where it acts	What it changes
Agent constants	Injected into v5.0's settings, so the policy and engine see them	Action fractions (10/20/30% of width), uncertainty multipliers, tier thresholds, novelty dampener
Calibration layer	Coach-owned, applied after v5.0's engine	Per-horizon centre gain λ and width scale ω . $\lambda=1$, $\omega=1$ reproduces v5.0 bit-for-bit

The layer exists because the highest-leverage lever — *base* interval width — has no in-engine control. v5.0's six uncertainty multipliers only fire when the LLM asks for a width change, and a neutral decision bypasses them entirely. Owning that lever in the coach keeps the agent unmodified.

Both levers, measured on a real record

Run 2026-03-03 at h=21, where the agent proposed `large_up` and cleared Tier 3:

Calibration	P50	Overlay	P10–P90 width
v001 (identity — what shipped)	72.964	+3.583	15.527
Layer width scale $\omega = 1.5$	72.964	+3.583	23.290
Agent constant: large fraction 0.30 → 0.20	71.770	+2.389	15.527

Width scales with the centre untouched; the agent constant scales the overlay by exactly 0.20/0.30. The two levers are independent, which is what lets the gate attribute a change to one hypothesis at a time.

The fidelity check

The check that keeps the coach honest

Replaying a record under the calibration that actually produced it must reproduce its recorded quantiles **exactly**, or the harness raises and stops. All five records currently reproduce bit-for-bit — max absolute error **0.0**, not merely within tolerance. That includes one record whose settings were reconstructed after the fact, which independently confirms the reconstruction was correct.

6 | SCORING

Three metrics, each answering a different question

The repo scored forecasts with CRPS only. CRPS is a reasonable summary and the wrong primary instrument for calibration work: it blends everything into one number, so it cannot tell you *which part* of the distribution was wrong. The coach's entire job is deciding whether to move the centre or the width, so it needs a loss that decomposes by quantile level.

Metric	What it answers	Role
Pinball loss	How wrong was each quantile, separately? Minimised only by the true quantile.	Primary. New to this repo — no quantile-loss function existed.
CRPS	One number per forecast, for comparability with existing repo results.	Reported. Both the quantile-integrated form and the framework's ensemble approximation.
80% coverage	Did the outcome land inside P10–P90? Target 80%.	R1's actual target. A rate, so it needs accumulated runs.

Effective sample size is reported, and it is brutal

Horizons overlap, so 60 daily runs at h=21 carry roughly three genuinely independent observations. Every scoreboard states the effective count next to the raw one, so $n=36$ never implies a precision that is not there. Today the effective count is **0.7**.

7 | THE LOCKED GATE

The only component allowed to say a change is an improvement

Everything else in the coach measures. This decides. The separation is deliberate: a scoreboard that also gets to declare victory will eventually be tuned until it does, and the failure is silent because every intermediate number still looks correct.

Nine conditions, all of which must hold. Six concern *what the corpus is*; three concern *what the numbers say*. The ordering matters — the statistical tests run last, because a p-value computed over a contaminated or pooled corpus is worse than no p-value: it launders the contamination into something that looks like evidence.

#	Condition	Why it exists
1	Provenance clean	Only live_forward records. Replayed history is not evidence.
2	Single prompt version	A different prompt is a different agent.
3	Single package fingerprint	If v5.0 changed underneath, old constants do not transfer.
4	Sufficient origins	Counted in origins, never in (origin, horizon) rows. Needs 12.
5	Holdout respected	Judged only on origins after what the candidate was fitted on. Needs 6.
6	Paired on identical days	Both variants scored on exactly the same keys.
7	Single tunable	One hypothesis per cycle, or the result cannot be attributed.
8	Effect size sufficient	Detectable is not the same as worth applying. Floor 3%.
9	Bootstrap significant	One-sided, resampling origins. $\alpha = 0.10$.

Why the bootstrap resamples origins, not rows

The three horizons of one run share an assessment, a research packet, a market state and overlapping outcome windows. They are close to one observation, not three. Resampling (origin, horizon) rows would treat them as independent, shrink the standard error by roughly $\sqrt{3}$, and manufacture significance out of correlation. So a resample draws whole origins and carries all of their horizons with it. There is a test asserting that the naive version looks *more* significant than the blocked one on identical data — the overclaim this exists to prevent.

Shrinkage, and what a pass actually means

A fitted value is a point estimate from a small, autocorrelated sample; its sampling error is large relative to the change being proposed, so applying it in full systematically over-corrects. A passed candidate is therefore shrunk halfway toward the incumbent before it is proposed — a little bias for a large reduction in the variance of what reaches a live forecast.

Passing is not applying

A passed verdict produces a *proposal* in a review queue, carrying the diff, the lineage, the statistics and the shrunk value. A human approves it, or it does not happen. Nothing in the coach writes a calibration version on its own.

8 | READING A RESULT

A reviewer checklist for the scoreboard

One command produces the whole picture in about two seconds, with no LLM and no network: `uv run python -m energy_oil_forecasting.cfm_coach.report`. Read it in this order.

Step	What to check	What good looks like
1. Fidelity	The drift banner at the top	All records replay exactly. Any drift invalidates every fit below it.
2. Corpus	Records, and how many are fitting-eligible	The fitting-eligible count is the one that matters.
3. Resolution	Resolved vs pending, and the price-data frontier	Pending is normal and expected; most of the corpus is unresolved most of the time.
4. Fitting band	The live_forward-only section	This is the only band that is evidence. Today it is empty and says so.
5. Variants	ensemble vs agent vs current	'current' equals 'agent' exactly while v001 is in force — proof the baseline is a true no-op.
6. Coverage	cover80 against the 80% target	R1's headline. Currently 0%.
7. Effective n	The independent-observation count	Governs what may be concluded. Compare it against the raw n.

The scoreboard today

Variant	n	Pinball	CRPS	MAE	Cover 80%	Width
agent (as published)	12	11.5434	21.4171	25.58	0.0%	11.08
current (v001)	12	11.5434	21.4171	25.58	0.0%	11.08
ensemble (no overlay)	12	12.3681	22.8239	26.93	0.0%	9.74

Do not over-interpret this table

All twelve resolved rows come from two March cutoffs that were **replayed**, not run live. The overlay appears to beat the raw ensemble by 6.7% on pinball. That is not evidence the overlay works: it is two cutoffs, re-executed against a web that already knew what happened, with roughly **0.7** effective independent observations. The report bands it separately for exactly this reason.

9 | WHAT WE HAVE LEARNED SO FAR

Five findings, four of them uncomfortable

The coach has not yet calibrated anything — it cannot, until the corpus exists. What it has already done is measure things nobody had measured, and four of those measurements contradicted assumptions the project was built on. That is the value delivered to date, and it is worth stating plainly.

1. The only prior evidence for the overlay was invalid

Both pre-existing audit files were issued 164 days after their cutoff. The '+10.9% overlay improvement on 6 of 6 horizons' measured early in this project comes entirely from those two replayed runs. It is **not** evidence that the overlay helps live. This is the finding that motivated making provenance a first-class, enforced field.

2. The trust tier does not discriminate

The trust reporter returns high for all three original records — including the two whose forecasts were off by \$27–\$35 with zero interval coverage. That is R4's retirement condition firing before a single forecast resolved. Neither sub-check caught March 2026: the out-of-distribution flag put that day at the 87.9th percentile of neighbour distance against a 99th-percentile threshold, and the five nearest historical analogues implied widths of $1.5 \times 0.9 \times 1.1 \times$ the ensemble against realized moves of +\$12/+\$25/+\$29.

The decision taken

Hold all of it. Nothing trust-related is wired into the daily path until M1 can rank it against realized error. The code stays in the tree, unused. A trust tier that does not separate outcomes is worse than none, because it is believed.

3. Citation bookkeeping — not market judgment — dominates run-to-run variance

The same cutoff was run twice, same code, same model, about fifteen hours apart. On 2026-03-03 the agent formed the **identical** view both times: same actions on all three horizons, same confidence, same novelty. The published output went from **+\$3.58 to +\$0.00**.

Cutoff	Issued	LLM proposed (h=21)	Policy gave	Why
2026-03-03	06:34	large_up	large_up, tier strong	—
2026-03-03	21:30	large_up	no_change , tier none	Cited claims failed the source-subset contract
2026-03-02	23:08	moderate_up	moderate_up, corroborated	11 resolved publishers
2026-03-02	21:25	large_up	small_up, limited	Only 5 resolved publishers

The second run wired its claim → summary → source citations in a way the evidence policy rejected. The policy behaved exactly as specified; the variance is upstream, in how claims are assembled. Three consequences, and they are large:

- Calibrating the overlay means fitting a signal that is being **randomly zeroed by an unrelated failure mode**. Any centre gain fitted across such runs is biased toward zero with inflated variance, and 40 records will not be enough if a meaningful share are nullified.
- The highest-leverage fix is **not a calibration constant**. It is the claim-building step.

- It is measurable **immediately**, with no waiting for horizons to resolve.

This was invisible before, because nothing had ever compared two runs of the same cutoff. Section 11 treats it as priority two, and explains why the timing is urgent.

4. The market-state signals are weaker than assumed

Measured over 4,679 historical business days, Spearman rank correlation against the absolute forward move:

Signal	move 5b	move 10b	move 21b
vix_level_11b	0.255	0.231	0.236
realized_volatility_21b	0.175	0.144	0.120
drawdown_63b	-0.174	-0.133	-0.109
return_1b	0.103	0.070	0.074
oil_curve_contango_11b	-0.086	-0.087	-0.097
jump_zscore_63b	0.045	0.018	0.020

- **Jump z-score has essentially no predictive power.** This kills the earlier hypothesis that the overlay's reach should scale with market abnormality — a hypothesis built on the 2.93 jump z-score observed on 2026-03-03. A large jump today does not predict a large move next.
- **The best state signal is VIX, and it is still weak** ($\rho \approx 0.24$). No state-based trust tier will discriminate strongly.
- **The usable effect is in the tail, not the median.** From the calmest to the most violent volatility quintile, the median 21-day absolute move rises only \$4.11 → \$5.40 — but the 90th percentile rises \$9.21 → **\$16.07**.

The leading calibration hypothesis, and it is now testable end to end

Width should scale with volatility. The centre should not. This serves R1 directly, it is supported by 4,679 days of history rather than by a handful of runs, and the harness can now measure whether it actually helps.

5. The replay is exact, and the baseline is a true no-op

The one unambiguously good finding. All five records reproduce their recorded quantiles bit-for-bit under their own calibration, and the scoreboard's 'current' column matches the 'agent' column to the digit. That means the coach's model of v5.0 is faithful, and that v001 changes nothing — so any future difference is attributable to a deliberate calibration rather than to drift.

10 | ROADMAP

What is built, what is designed, what is deferred

Stage	Contents	State
1a — Recorder	Run record schema, store, ledger, daily entry point, backfill	Complete and verified
1b — Trust report and M3a	State diagnostics, 4,700-case analogue base, trust reporter	Built; deliberately not wired in pending R4
1c — M1 harness	Replay, scoring, outcomes, scoreboard, locked gate	Built and tested ; idle until the corpus arrives
1c — Workflow B	Candidate generation, proposal builder, review queue, cycle runner	Not built
1d — M2 memory	Self-audit of novelty, confidence, persistence and claim-type calibration, rendered into the prompt as a compact table	Designed
2 — M3b	Analogue injection into the prompt	Deferred

M2: what the agent would learn about itself

The agent produces judgments, not calculations, so its self-audit is a judgment audit. All four targets reach the forecast through a *typed* channel — the LLM's categorical outputs are load-bearing inputs to a policy Python still owns — so this is not prose-to-number leakage.

Target	The question it answers
Novelty calibration	Highest leverage: novelty both gates eligibility and applies the 0.5× dampener. 'When I called this likely new, did the price move like new information?'
Confidence calibration	Do 0.9-confidence calls actually beat 0.6-confidence calls? The tiers assume so; nothing has checked.
Persistence	The persistence profile is collected every horizon and used by nothing. Did 'persistent' calls hold at h=21?
Claim-type efficacy	The Tier 3 gate bets that physical evidence beats market-reaction evidence. Does it?

The trap that shapes the whole schedule

Turning on M2 changes what the LLM says, so every earlier run came from a different agent and pooling across prompt versions when fitting is biased. The discipline: prompt version is required on every record from day one; the prompt is frozen while M1 establishes a baseline; and M2 is introduced as an explicit new agent version, evaluated by M1's paired harness. The same logic defers M3b — the cost of analogue retrieval is not the search, it is putting results into the prompt.

11 | NEXT STEPS

In priority order, highest value first

The ordering below is by *value per unit of cost and delay*, not by dependency alone. Two items are urgent for reasons of timing rather than effort: they get much more expensive the longer the corpus grows.

#	Action	Why now	Cost	Needs a decision?
P1	Keep the daily job running, uninterrupted	Everything is blocked on corpus, and a missed weekday is evidence that cannot be recovered — backfilling re-searches today's web. Confirm it fires each morning.	Zero — already scheduled	No
P2	Fix the claim-building instability (finding 3)	It randomly zeroes the very signal M1 exists to fit. Fixing it is a prompt change , which creates a new prompt version and fragments the corpus — so it is cheapest to do now , at one live record, and roughly forty times more expensive in two months.	Small; measurable within a day	Yes — see below
P3	Run the deterministic base corpus	The ensemble alone over the 51+18 existing origins, no LLM, minutes of compute. Yields ~200 scored base-layer forecasts <i>immediately</i> , which is the fastest available evidence on how wrong the interval widths are.	Minutes	No
P4	Widen base intervals as a declared prior (v002)	Coverage is 0 of 12 and 4,679 days of history say the tails are badly under-served. Declared with no fitted period, so all future data judges it out-of-sample and honestly.	Low	Yes — it changes live output
P5	Build workflow B (proposal + cycle runner)	Needed before the first calibration cycle can run, but not before the corpus exists. No reason to rush it, and no reason to leave it to the last week.	Moderate	No
P6	Run the contamination test	Replay the 2026-08-14 cutoff in a month or two and diff the assessment against the live original. Cheap, and it settles a design question the March comparison could not.	One run	No
P7	First real calibration cycle	At roughly 40 live_forward records — about two months of daily running. The first tunable should be interval width; the centre gain second.	—	Yes — approve or reject
P8	M2 self-audit memory	Only once M1 has a baseline that can A/B it. Introducing it earlier means there is no instrument to tell whether it helped.	Moderate	Yes — new prompt version
P9	M3b analogue injection	Stage 2. Same prompt-version cost as M2, with less expected value until novelty calibration is understood.	Moderate	Yes

The one decision that needs the v5.0 owner

P2 requires modifying CFM Agent v5.0, and the coach cannot make that call

The claim-building instability lives in `cfm_agent_v_5_0/skills/claim-building/SKILL.md`, which is **inside** the agent package and covered by its manifest. The coach's hard rule is that it never modifies v5.0 — so this is a change only the agent's owner can authorise, and it would change the package fingerprint, making records before and after it formally a different system for fitting purposes.

The recommendation: do it deliberately and early, as an explicit version bump of the agent, while the corpus is one record. The alternative — leaving it — means M1 spends two months fitting a signal that is being randomly set to zero by something unrelated to market judgment.

What good looks like in three months

- Roughly 60 `live_forward` records, of which ~40 have at least one resolved horizon.
- 80% interval coverage measured rather than assumed — and if it is still near zero, a width calibration that has passed the gate and been approved.
- A definitive answer to whether the LLM overlay beats the raw ensemble on *live* runs. This has never been measured and is the single most valuable number the project can produce.
- The trust tier either validated against realized error, or formally retired under R4.
- The claim-building variance either fixed and confirmed by re-running the same cutoff twice, or quantified so calibration can account for it.

12 | PRACTICAL GUIDANCE

Strengths, limitations, and shared terminology

Key strengths

- The agent is genuinely untouched: additive package, injected settings, verified 43-file manifest.
- Replay is exact rather than approximate, and that claim is re-checked on every cycle.
- Contaminated evidence is excluded by construction, not by remembering to.
- One component, and only one, may declare an improvement — behind nine conditions.
- The statistics respect the dependence structure: origin-blocked bootstrap, paired days, reported effective sample size.
- Nothing reaches a live forecast without a human approving it.

Important limitations

- **The corpus is one record.** Everything the harness can conclude today, it concludes about replayed history, which is not evidence.
- The trust report does not yet discriminate and is held back for that reason.
- Analogue dispersion is a plausibility check, not a measured error distribution. It can say an interval looks wrong; only accumulated coverage says by how much.
- The quantile-integrated CRPS spans the 5th to 95th percentile, so it is a consistent lower bound rather than an absolute CRPS. Fine for paired comparison; not for quoting externally.
- Calibrating the machinery cannot fix a mis-specified ensemble. Improving the underlying numerical models is real work and a separate project.
- Run-to-run variance in the agent's citation assembly is currently unmodelled and will inflate the variance of any fit until it is addressed.

Glossary

Term	Meaning
Origin	One forecast cutoff. The unit the bootstrap resamples, and the unit sample size is counted in.
Provenance	Whether a run was issued live or re-executed at a past cutoff. Only live_forward is fitting evidence.
Fidelity	Whether replaying a record under its own calibration reproduces its recorded output exactly.
Pinball loss	Quantile loss. Minimised only by the true quantile, and decomposable by level.
Coverage	The fraction of outcomes landing inside P10–P90. Target 80%.
Calibration layer	The coach-owned post-engine transform: per-horizon centre gain and width scale.
Calibration version	A dated, immutable JSON document of constants. v001 is a true no-op.
Holdout	Origins after the candidate's declared fitting period. The only ones the gate judges on.
Shrinkage	Moving a fitted value partway back toward the incumbent before proposing it.
Effective n	Independent observations, after accounting for overlapping horizons. Much smaller than the row count.

Bottom line

CFM Agent v5.0 makes disciplined forecasts and has never been told whether they were any good. CFM Coach is the instrument that will tell it — recording every run so it can be replayed exactly, scoring against what actually happened, and holding any proposed change behind a deliberately conservative gate. It has already found four things that were being assumed incorrectly. What it cannot do is hurry: the corpus accrues one weekday at a time, and the discipline that makes the eventual answer trustworthy is the same discipline that makes it slow.